CHAPTER III

METHODOLOGY

This chapter outlines the methodology used for the study "MEDFLOW PH: An Unsupervised Clustering Analysis of Medicine Procurement and Distribution Data in Philippine Public Health Facilities." The methodology follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, which is widely used for data mining projects, especially those involving large datasets like the one in this study. CRISP-DM provides a structured and systematic approach for addressing data mining problems, from understanding the problem to generating actionable insights based on data analysis. This chapter will detail the six phases of the CRISP-DM methodology and how each phase contributes to achieving the study's objectives.

The CRISP-DM methodology includes the following phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. Each phase is integral to the process, ensuring the successful analysis of medicine procurement and distribution data through clustering techniques. The study uses unsupervised clustering methods, particularly k-means and DBSCAN, to extract patterns from the data and identify inefficiencies in the supply chain.

CRISP-DM METHODOLOGY

The CRISP-DM methodology is the most widely adopted approach in the field of data mining. It is designed to provide a structured process for analyzing data and transforming it into useful insights. The six phases of CRISP-DM allow for an iterative approach to solving complex problems, ensuring that each step builds upon the previous one to derive actionable results.

In the case of this study, CRISP-DM helps ensure that the clustering analysis of medicine procurement and distribution data is systematic, providing clear and interpretable insights. Each phase is carefully tailored to the unique needs of this research, where the primary objective is to identify inefficiencies in medicine distribution and procurement across Philippine public health facilities.

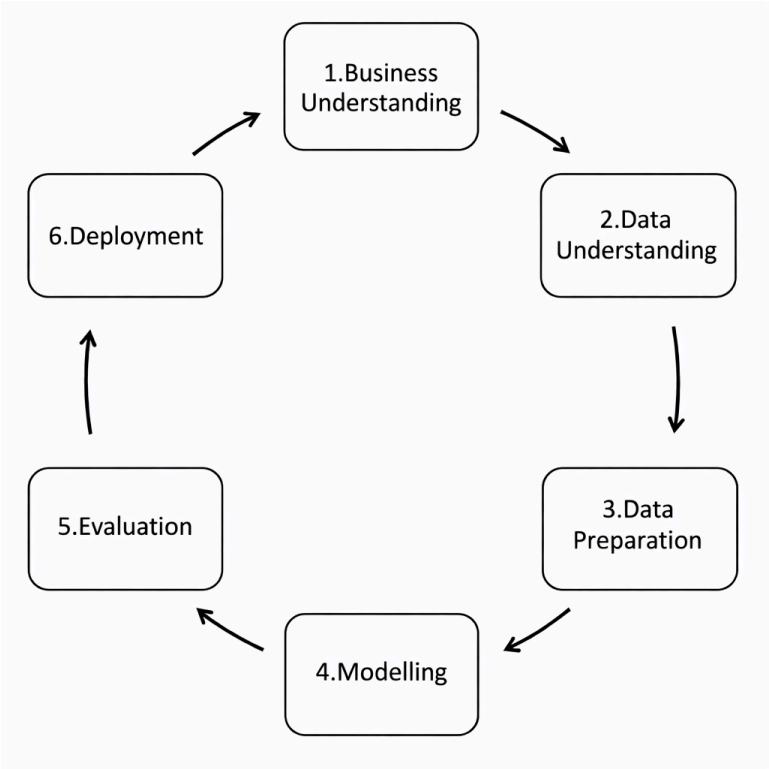


Figure 2. CRISP-DM framework

BUSINESS UNDERSTANDING

The Business Understanding phase is the first and most crucial step in the data mining process. It defines the objectives of the study, aligns the data mining process with real-world goals, and ensures that the results will provide actionable insights.

The primary goal of the MEDFLOW PH study is to analyze the medicine procurement and distribution data from public health facilities in the Philippines to identify inefficiencies in the supply chain. Issues like stockouts, overstocking, and ineffective distribution practices are common, especially in remote or underserved regions, which affect the availability of essential medicines. By applying unsupervised clustering analysis, this study seeks to uncover these inefficiencies, providing recommendations for policy improvements and more effective medicine supply chain management.

Key activities during this phase include:

- Problem Definition: The inefficiencies in the procurement and distribution processes, such as uneven medicine availability and inconsistent distribution, are defined as the core business problem.
- Setting Objectives: The aim is to identify clusters of public health facilities that exhibit similar procurement and distribution patterns. This will allow stakeholders to address inefficiencies specific to each cluster.

- Stakeholder Engagement: The study will collaborate with key stakeholders such as public health administrators, policy makers, and supply chain managers to ensure the analysis addresses real-world concerns.

The expected outcome of this phase is a clear problem statement and a set of objectives that will guide the data mining process, ensuring that the system's results can directly inform decisions and improve the procurement and distribution processes.

DATA UNDERSTANDING

The Data Understanding phase is where the data is collected, explored, and analyzed to identify any issues or patterns that could impact the clustering analysis. The data for this study will be collected from publicly available sources such as FOI.gov and regional health department records. The data will include medicine types, quantities procured, quantities distributed, facilities, and regional information.

Key activities during this phase include:

- Data Collection: Gathering procurement and distribution records from government databases and health department records. This will provide the data needed for analysis.
- Exploratory Data Analysis (EDA): Conducting initial analysis to understand the structure of the data and assess its quality. This includes identifying:
    - Missing values: Any missing or incomplete records that may require imputation or removal.

  - Outliers: Recognizing extreme values that could influence the clustering results.
  - Variable distributions: Evaluating the distribution of key variables such as quantities procured and distribution volumes.
- Understanding Variables: Identifying the most important variables for clustering. These might include medicine types, regions, quantities procured, and distribution efficiency.

The expected output of this phase is a comprehensive understanding of the data, including its structure and quality, and identifying any potential challenges to the analysis. The data will then be cleaned and transformed for the next phase.

DATA PREPARATION

The Data Preparation phase involves cleaning, transforming, and organizing the data into a format suitable for clustering analysis. This phase is crucial as the quality of the prepared data directly impacts the effectiveness of the clustering results.

Key activities during this phase include:

- Data Cleaning: Addressing missing values, errors, and inconsistencies. Missing data will be imputed where necessary, and problematic records (e.g., duplicates) will be removed.

- Feature Engineering: Creating new variables that will be helpful for clustering. For example, calculating a demand-to-supply ratio could help identify regions with procurement or distribution imbalances.
- Data Transformation: Normalizing numerical variables such as procurement quantities and distribution volumes so that no one variable disproportionately affects the clustering results.
- Categorical Data Encoding: Converting categorical data, such as medicine types and regions, into numerical formats using one-hot encoding or label encoding for compatibility with clustering algorithms.
- Feature Selection: Identifying the most relevant features to use in the clustering process to avoid unnecessary complexity in the model.

The output of this phase is a cleaned, preprocessed dataset that is ready for the modeling phase, with features that are optimized for the clustering process.

MODELING

The Modeling phase is where the core data mining work happens. This phase involves applying unsupervised clustering algorithms to the prepared data to identify patterns and group similar data points together.

Key activities during this phase include:

- Algorithm Selection: We will apply k-means and DBSCAN clustering algorithms. k-means is used to partition the data into a pre-defined number of clusters, while DBSCAN is more flexible, capable of detecting clusters of varying shapes and handling noise (outliers).

- Parameter Tuning: For k-means, the Elbow Method or Silhouette Score will be used to determine the optimal number of clusters (k). For DBSCAN, the parameters epsilon (neighborhood size) and min_samples (minimum number of points in a cluster) will be adjusted for optimal performance.
- Model Training: The clustering algorithms will be trained on the prepared data, grouping similar facilities or medicines based on procurement and distribution patterns.
- Cluster Assignment: Once the model is trained, each data point (i.e., each facility or medicine type) will be assigned to a cluster.

The expected output of this phase is the identification of clusters within the data, such as facilities with similar procurement patterns or regions with similar medicine distribution needs.

EVALUATION

The Evaluation phase assesses the quality of the clusters and determines whether the clustering model provides useful insights. It ensures that the clusters generated are meaningful and relevant to the business objectives.

Key activities include:

- Cluster Validation: Using metrics such as the Silhouette Score and Inertia, we will evaluate the quality of the clusters. A higher Silhouette Score indicates well-separated and cohesive clusters, while lower Inertia suggests better-defined clusters.

- Cluster Interpretation: We will analyze the characteristics of the clusters to determine whether they align with the business objectives. For example, clusters showing frequent stockouts or overstocking could provide insights for improving medicine procurement strategies.

The expected output of this phase is an evaluation report that highlights the quality of the clusters and provides actionable insights into the medicine supply chain.

DEPLOYMENT

The Deployment phase ensures that the results of the clustering analysis are accessible and useful to stakeholders, such as public health administrators and policy makers. This phase focuses on making the insights actionable.

Key activities during this phase include:

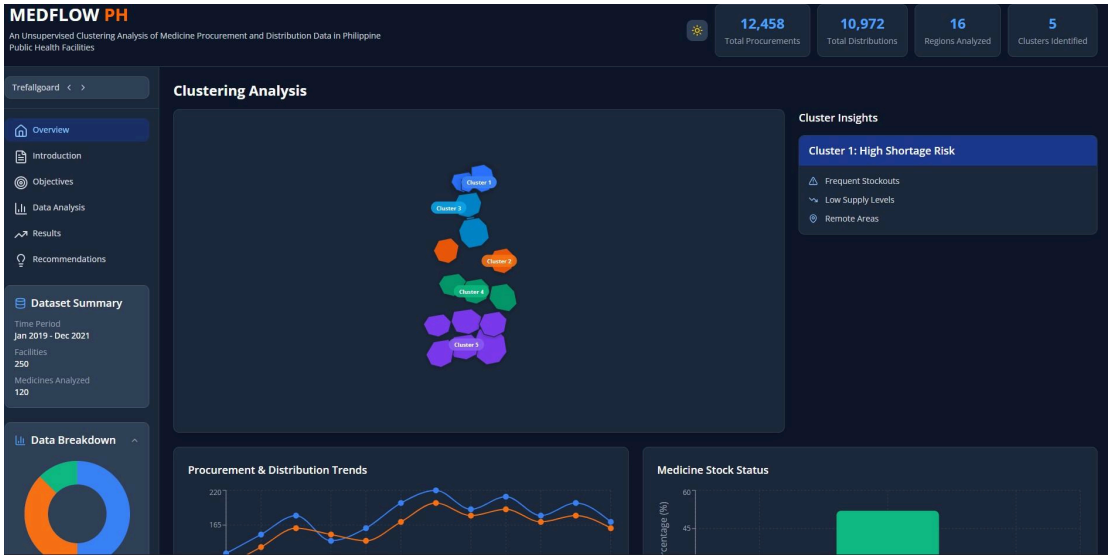
- Report Generation: Visualizations such as heat maps, bar charts, and scatter plots will be created to display clustering results and highlight key insights, such as regions with medicine shortages or overstocking.
- Actionable Insights: Based on the clusters, recommendations will be provided for improving medicine procurement and distribution, such as redistributing overstocked medicines or increasing procurement in stockout-prone areas.

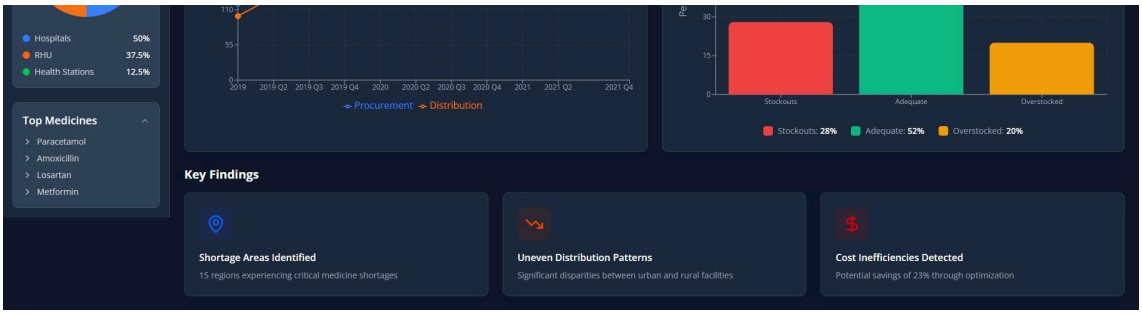
The expected output of this phase is a set of interactive reports or visualizations that provide stakeholders with actionable insights.

Initial Layout and Designs

The initial layout and design of the MEDFLOW PH system are developed to support exploratory analysis, interpretation, and decision making based on unsupervised clustering results derived from aggregated and non personal medicine procurement and distribution data. The design prioritizes clarity, consistency, and analytical relevance to ensure that complex clustering outputs are presented in an accessible and interpretable manner for public health administrators, logistics managers, and data analysts.

The system is implemented as a web based dashboard interface that integrates summary indicators, clustering visualizations, trend analysis, and interpretive insights within a unified analytical environment. The overall layout is structured to guide users progressively from high level system summaries to more detailed analytical views, thereby supporting both overview monitoring and focused investigation of procurement and distribution patterns.





The interface is organized into three main structural components, namely a header section, a navigation panel, and a central content area. The header section, positioned at the top of the interface, displays the system name MEDFLOW PH and a brief description identifying it as an unsupervised clustering analysis platform for medicine procurement and distribution in Philippine public health facilities. This section also presents key system indicators, including the total number of records analyzed, the number of health facilities and regions covered, and the number of clusters identified. These indicators provide immediate contextual information regarding the scale and coverage of the analysis.

The navigation panel is located on the left side of the interface and serves as the primary access point for the system's analytical functions. It contains menu options that allow users to navigate between the overview, clustering analysis, dataset summary, data breakdown, and medicine level insights. This structure supports logical navigation across analytical views while maintaining continuity in the user's analytical workflow.

The central content area integrates the clustering analysis view with supporting analytical charts and data summaries to provide a comprehensive interpretation of medicine supply patterns. At the core of this area is the

clustering visualization, which graphically represents the results of unsupervised clustering using K-means and DBSCAN algorithms. Each visual element represents a grouped entity, such as a health facility or medicine category, based on similarities in procurement and distribution behavior. Clusters are visually distinguished through color coding, allowing users to assess separation, density, and relative proximity among groups. An adjacent cluster insights panel provides a concise interpretive summary of the selected cluster, highlighting dominant supply characteristics such as potential shortage risk, stable distribution behavior, or irregular procurement and distribution patterns.

Supporting analytical charts are presented within the same view to contextualize the clustering results using temporal and categorical perspectives. Trend charts display procurement and distribution volumes across reporting periods, enabling users to compare inflow and outflow patterns and to identify inconsistencies or fluctuations over time. Additional charts summarize inferred medicine stock status, presenting comparative views of adequate supply, low stock conditions, and potential shortage indicators derived from aggregated data. Proportional summaries further describe the composition of the dataset by facility type, medicine classification, and geographic coverage. The integration of clustering outputs with trend analysis and data summaries allows users to interpret unsupervised learning results within the broader operational context of medicine supply management.

The lower section of the interface is dedicated to the presentation of key findings and interpretive insights generated from the clustering analysis. These

findings are displayed as concise insight cards that summarize critical observations, such as the identification of shortage prone areas, uneven distribution patterns, or cost related inefficiencies. Each insight card includes a brief explanatory description that translates analytical results into actionable knowledge while avoiding excessive technical detail. This design supports the system's role as a decision support tool rather than a purely technical analytics platform.

The initial layout and design of the MEDFLOW PH system are closely aligned with the objectives of the study. By integrating unsupervised clustering results, temporal trends, dataset summaries, and interpretive insights within a cohesive web based interface, the system facilitates data driven monitoring, analysis, and planning related to medicine procurement and distribution. The design ensures that clustering outputs generated through K-means and DBSCAN are presented in a structured and meaningful manner that supports evidence-based decision making within Philippine public health facilities.

Use Case Diagram

The use case diagram illustrated in *Appendix A,* illustrates the interactions between various stakeholders and the MEDFLOW PH system in the context of medicine procurement and distribution analysis. The diagram is organized to show the horizontal relationships between the system's users and their respective use cases. Key stakeholders include the Public Health Administrator, Logistics Manager, Healthcare Facility, and Data Analyst, each of whom interacts

with the system to achieve specific objectives related to the clustering analysis of medicine data. The diagram highlights the following core use cases:

- The Public Health Administrator who analyzes procurement and distribution data, as well as implements policy decisions based on clustering insights.
- The Logistics Manager who monitors regional distribution clusters and adjusts the distribution strategy to optimize medicine availability.
- The Healthcare Facility which views insights on medicine availability based on clustering results.
- The Data Analyst responsible for monitoring and reporting clustering results, as well as generating the necessary data insights to support operational decision-making.

This use case diagram emphasizes the collaborative efforts among stakeholders to enhance medicine procurement and distribution through data-driven decision-making processes.

Activity Diagram

The activity diagram illustrated in *Appendix B* depicts the flow of tasks involved in the process of analyzing medicine procurement and distribution data using unsupervised clustering techniques, as outlined in the MEDFLOW PH study. The diagram begins with the collection of procurement and distribution data from various public health facilities. Once the data is collected, it undergoes

preprocessing, which includes cleaning and normalization to ensure consistency and accuracy.

Next, the appropriate clustering algorithm, such as K-means or DBSCAN, is selected and applied to the data. If the clustering result is valid, the analysis continues by interpreting the clusters and identifying potential inefficiencies or risks in medicine distribution, such as stockouts or overstocking. If the clustering results are not valid, the algorithm's parameters are adjusted or a different algorithm is applied.

The final output of this process is the generation of actionable insights that can guide policy decisions, distribution strategies, and other operational improvements to enhance medicine availability and reduce inefficiencies in the public health system.

Sequence Diagram

The sequence diagram illustrated in *Appendix C,* illustrates the interactions between the Data Analyst, the Data Source, and the Clustering Model in the process of performing the clustering analysis of medicine procurement and distribution data. The diagram shows the following sequence of events:

- The Data Analyst first requests procurement and distribution data from the Data Source, which returns the necessary data for further processing.

- After receiving the data, the Data Analyst preprocesses the data, which includes cleaning and transforming the data to ensure consistency and accuracy.
- The preprocessed data is then passed to the Clustering Model, where unsupervised clustering techniques, such as K-means or DBSCAN, are applied to identify hidden patterns or inefficiencies in the distribution process.
- Once the clustering process is complete, the Result Viewer displays the clustering results, which are then analyzed by the Data Analyst to identify distribution patterns and inefficiencies, such as stockouts or overstocking in different regions or facilities.

Component Diagram

The component diagram, illustrated in *Appendix D* represents the architecture of the MEDFLOW PH system, depicting the key components and their interactions. The system is divided into four main modules:

- *Data Sources*: This module includes the Procurement Data and Distribution Data components, which provide the raw data needed for analysis.
- *Data Processing*: The Data Cleaner prepares the data by cleaning and normalizing it, while the Feature Engineering component transforms the data to make it suitable for clustering analysis.

- *Modeling*: The K-means Algorithm and DBSCAN Algorithm are used to perform unsupervised clustering on the data, identifying patterns and inefficiencies in the medicine distribution system.
- *Result Generation*: The Cluster Viewer visualizes the clustering results, and the Insights Reporter generates actionable reports and recommendations based on the clustering patterns.

This diagram highlights how data flows through the system, from the collection of procurement and distribution data to the generation of insights that inform operational decisions and policy-making.

Deployment Diagram

The deployment diagram illustrated in *Appendix E*, illustrates the physical architecture of the MEDFLOW PH web based application, highlighting how system components are distributed across logical nodes and how they interact during execution. Unlike a desktop based system, MEDFLOW PH is implemented as a web application using the Flask framework and operates on a static dataset containing aggregated medicine procurement and distribution records covering the period from 2020 to 2025. As the dataset is fixed and no new data are added during deployment, a database server is not required within the scope of this study.

The system architecture consists of three main nodes, namely the Client Browser, the Web Application Server, and the Dataset File Repository.

The Client Browser node represents the end user environment. Stakeholders such as public health administrators, logistics managers, and data analysts access the MEDFLOW PH system through a standard web browser. The browser provides the presentation layer of the system, allowing users to view clustering results, interact with visualizations, and examine recommendations generated from the analysis. All computational tasks are handled on the server side, while the client device is responsible only for rendering the user interface and displaying results.

The Web Application Server node hosts the Flask application and contains the core analytical and processing components of the system. This node includes the Data Processing Module, which performs data loading, cleaning, transformation, encoding, and normalization of the aggregated procurement and distribution data. It also includes the Clustering Model, which executes the K-means and DBSCAN algorithms to identify patterns and groupings within the dataset. The server further hosts the Result Visualization and Recommendation Module, which generates cluster summaries, charts, and analytical insights that support interpretation and decision making related to medicine distribution in public health facilities.

The Dataset File Repository node stores the static data files used by the system. These files are maintained in file based formats such as CSV or Excel and contain aggregated and non personal medicine procurement and distribution records. During system execution, the Web Application Server retrieves the dataset from this repository for preprocessing and clustering analysis. Since the

data are historical and fixed within the study period, the repository functions as a read only data source rather than a dynamic database.

The deployment diagram illustrates the interaction between these nodes, with the Web Application Server accessing the dataset files from the repository, performing clustering analysis using K-means and DBSCAN, and transmitting the processed results to the Client Browser for visualization and interpretation. This deployment structure ensures a clear separation between user interaction, analytical computation, and data storage, supporting reliable and reproducible analysis within the defined scope of the study.

Data Flow Diagram

The Data Flow Diagram illustrated in *Appendix F*, illustrates the end-to-end analytical process used in the MEDFLOW PH study, showing how medicine procurement and distribution data are transformed into decision-support insights. The process begins with aggregated medicine procurement and medicine distribution data sourced from public health facilities, which serve as the primary inputs to the system. These datasets are consolidated during the data collection stage, aligned with the CRISP-DM framework, to ensure that all relevant operational variables are systematically gathered and structured. The collected data then undergo data preprocessing, which includes cleaning, normalization, and preparation to address inconsistencies, missing values, and scale differences. The resulting processed dataset is stored and subsequently used as input for clustering analysis, where

unsupervised machine learning techniques are applied to identify latent patterns and groupings within the data. The outputs of the clustering process are examined during the cluster interpretation stage, where meaningful patterns related to medicine distribution behavior and potential systemic risks are identified. These interpreted results are then translated into decision support outputs that inform public health stakeholders. Insights generated from the analysis feed back into the public health system, supporting data-driven planning, monitoring, and policy-oriented decision-making related to medicine procurement and distribution.

State Diagram

The state diagram illustrated in *Appendix G*, illustrates the step-by-step analytical workflow implemented in the MEDFLOW PH study, detailing how raw medicine procurement and distribution data are transformed into actionable insights through unsupervised clustering analysis. The workflow begins with the collection of aggregated and non-personal data from public health sources, after which the dataset undergoes preprocessing to ensure analytical readiness. This preprocessing stage includes cleaning missing or invalid values, normalizing numerical variables, encoding categorical attributes, and transforming the data into consistent formats. Feature selection is then performed to identify the most relevant variables that capture meaningful procurement and distribution behavior while reducing noise and dimensionality. Once the features are prepared, clustering analysis is conducted using both k-means and DBSCAN algorithms to

identify natural groupings and structural patterns within the data. The resulting clusters are evaluated and validated to assess their quality and coherence, followed by systematic interpretation to derive substantive meaning from the identified groupings. The workflow concludes with visualization and reporting, where clustered results are translated into graphs, summaries, and analytical outputs that support insight generation. These insights serve as inputs for data-driven decision-making related to medicine supply monitoring, planning, and resource allocation within public health systems.

Hardware and Software Requirements

The MEDFLOW PH system is implemented as a web based analytical application designed to process a fixed and non evolving dataset covering medicine procurement and distribution records. Since the system focuses on exploratory data analysis and unsupervised clustering using static data, it does not require continuous data ingestion or persistent transactional storage. The hardware and software requirements presented in this section are therefore tailored to support data preprocessing, clustering computation, visualization, and report generation within a controlled research environment.

Hardware Requirements

Hardware requirements cover the minimum and recommended computing specifications for running the MEDFLOW PH system in a local or hosted web server environment. The system performs data preprocessing, clustering model computation, and visual rendering of results. These tasks can be supported by

standard computing devices, although higher specifications improve performance when working with larger datasets, repeated clustering runs, and interactive visualizations.

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i5 or equivalent	Intel Core i7 or equivalent
Memory (RAM)	8 GB	16 GB or higher
Storage	256 GB SSD	512 GB SSD or higher
Display Resolution	1366 × 768	1920 × 1080 or higher
Operating System	Windows 10 (64 bit)	Windows 11 (64 bit)
Network	Stable local connection for access to the web application	Stable local or broadband connection

Table 1. Hardware Requirements

The minimum configuration supports standard preprocessing and clustering execution using K-means and DBSCAN for a fixed dataset. The recommended configuration improves responsiveness, especially during iterative parameter tuning, metric computation, and visualization rendering.

Software Requirements

Software requirements describe the development and runtime environment needed to implement and execute the MEDFLOW PH system. The

system is implemented as a web application using Flask. Data preprocessing and clustering analysis are performed using a Python based analytics stack. Since the dataset is static and limited to a defined period, the system reads the dataset from file based sources such as CSV or Excel and processes it within the server environment during runtime. Persistent database storage is not required within the defined scope of this study.

Software Category	Specification and Purpose
Operating System	Microsoft Windows 10 or Windows 11
Programming Language	Python version 3.10 or later
Web Framework	Flask for web application routing and server side processing
Data Handling Libraries	Pandas and NumPy for data cleaning, transformation, and feature engineering
Machine Learning Libraries	Scikit learn for K-means clustering, DBSCAN clustering, and validation metrics
Visualization Libraries	Matplotlib and Plotly for charts, cluster summaries, and interactive views
Dataset Format	CSV or Excel files containing aggregated procurement and distribution records
Development Environment	Visual Studio Code or equivalent Python integrated development environment

Table 2. Software Requirements

This software stack supports the full analytical pipeline required in the study, including preprocessing, clustering execution using K-means and DBSCAN, evaluation using internal validation metrics, and visualization for interpretation and reporting. The use of Flask enables the presentation of clustering outputs through a web interface that supports filtering, viewing cluster profiles, and exporting analytical summaries.

Test Cases

Testing is conducted to verify that the MEDFLOW PH web based system performs its intended analytical functions correctly and consistently. The test cases focus on validating the end to end workflow of the system, from dataset loading and preprocessing to clustering execution, evaluation, visualization, and report generation. The purpose of testing in this study is not software performance benchmarking but functional verification of the analytical processes that support unsupervised clustering analysis of medicine procurement and distribution data.

The test cases are designed to ensure that the system correctly handles valid and invalid inputs, produces meaningful clustering results using K-means and DBSCAN, and presents outputs in a clear and interpretable manner suitable for decision support. Table 3 summarizes the test cases applied in the study.

Test Case ID	Test Objective	Input Conditions	Process	Expected Result
01	Verify dashboard loads with dataset summary	Valid dataset loaded	Open dashboard page	System displays dataset coverage, counts, and key summary statistics
02	Verify filtering by region and facility type	Dataset includes multiple regions and facility types	Apply filters and refresh view	Dashboard visualizations and tables update correctly based on selected filters
03	Verify cluster visualization rendering	Clustering results available	Open cluster visualization page	Cluster distribution charts and cluster summaries are displayed without rendering errors
04	Verify facility level drill down	Selected facility exists in dataset	Search and open facility details	System shows facility specific procurement and distribution trends and assigned cluster label
05	Verify export of results	Clustering results available	Export report as CSV or PDF	Exported file contains correct cluster labels, summaries, and selected filters, and download completes successfully

Table 3. Test Cases